Logical Equivalence Laws

Logical Equivalence Laws

Name of the Law	Law	Law
Commutative	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity	$p \wedge T \equiv p$	$p \vee F \equiv p$
Negation	$p \vee \sim p \equiv T$	$p \wedge \sim p \equiv F$
Double Negation	$\sim(\sim p) \equiv p$	
Idempotent	$p \wedge p \equiv p$	$p \vee p \equiv p$
Universal Bound	$p \vee T \equiv T$	$p \wedge F \equiv F$
DeMorgan’s	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
Absorption	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negations of T & F	$\sim T \equiv F$	$\sim F \equiv T$

Proof of Commutative Law

Name of the Law	Law	Law
Commutative	$p \land q \equiv q \land p$	$p \lor q \equiv q \lor p$

p	q	$p \land q$	$q \land p$	$p \lor q$	$q \lor p$
F	F	F	F	F	F
F	T	F	F	T	T
T	F	F	F	T	T
T	T	T	T	T	T

Proof of DeMorgan's Law

Name of the Law	Law	Law
DeMorgan's	∼(p ∧ q) ≡ ∼p ∨ ∼q	∼(p ∨ q) ≡∼p ∧ ∼q

p	q	∼p	∼ q	p ∧ q	∼(p ∧ q)	∼p ∨ ∼q
F	F	T	T	F	T	T
F	T	T	F	F	T	T
T	F	F	T	F	T	T
T	T	F	F	T	F	F

∴∼ (p ∧ q) ≡ ∼ p ∨ ∼ q [Proved]

p	q	∼p	∼ q	p ∨ q	∼(p ∨ q)	∼p ∧ ∼q
F	F	T	T	F	T	T
F	T	T	F	T	F	F
T	F	F	T	T	F	F
T	T	F	F	T	F	F

∴∼ (p ∨ q) ≡∼ p ∧ ∼ q [Proved]

Assignment 1 – Submission Deadline: Before Mid-term Exam

- Prove all of the Logical Equivalence Laws